

unbound: chroot hardcoded to /var/unbound prevents outbound DNS forwarding

Component: Services: Unbound DNS

Version: OPNsense 26.1.5 (confirmed present in master as of 2026-03-27)

Severity: High — complete forwarding failure in all forwarding-mode deployments

File: src/etc/inc/plugins.inc.d/unbound.inc

Line: 341 (generated unbound.conf template)

Summary

In `unbound_generate_config()`, the generated `unbound.conf` always contains `'chroot: /var/unbound'`. The `/var/unbound` chroot environment lacks the network device nodes required for Unbound to open outbound UDP sockets and reach upstream resolvers. When DNS forwarding is configured, all forward queries fail silently — Unbound receives the query, attempts to forward, but cannot send packets, resulting in `SERVFAIL` or `NXDOMAIN` responses to clients.

Environment

OPNsense 26.1.5 (FreeBSD 14) | Unbound 1.24.2 | DNS Query Forwarding enabled | Upstream resolvers: 1.1.1.1, 8.8.8.8 | Unbound bound to internal interface IP (10.10.11.1)

Steps to Reproduce

1. Configure Unbound with Query Forwarding enabled
2. Verify chroot: `/var/unbound` in `/var/unbound/unbound.conf`
3. Attempt DNS resolution: `host google.com 127.0.0.1`
4. Run packet capture: `tcpdump -i re0 -n udp port 53`
5. Observe 0 outbound packets despite `NXDOMAIN` response

Expected Result

Unbound forwards queries to upstream resolvers. Packet capture shows outbound UDP/53 traffic to 1.1.1.1.

Actual Result

```
$ host google.com 127.0.0.1
Host google.com not found: 3(NXDOMAIN)
```

Packet capture: 0 packets captured on WAN interface.

```
unbound-control lookup google.com:
1.1.1.1    rto 120000 msec, tA 3, tAAAA 3, tother 3
(Unbound has recorded repeated timeouts – it attempts but fails to reach forwarders)
```

Root Cause

Line 341 generates:

```
chroot: /var/unbound
```

The chroot at /var/unbound is a restricted filesystem. While it contains /dev via devfs and Python lib

Impact

Any OPNsense 26.1.5 deployment using Unbound in forwarding mode with chroot active will silently fail to resolve external domains. The failure mode is non-obvious: Unbound starts successfully, binds correctly, accepts queries, but returns NXDOMAIN. No error is logged at default verbosity.

Fix

```
# Before (generated unbound.conf, line 341):
chroot: /var/unbound
```

```
# After:
chroot: ""
```

In unbound_generate_config(), change the hardcoded chroot path to an empty string.

Validation

```
$ grep 'chroot' /var/unbound/unbound.conf
chroot: ""
```

```
$ tcpdump -i re0 -n udp port 53 &
$ host google.com 127.0.0.1
192.168.1.248.47815 > 1.1.1.1.53: A? google.com.
1.1.1.1.53 > 192.168.1.248.47815: 6/0/1 A 142.250.114.139 ...
```

Outbound forwarding packets visible on WAN interface.

Suggested Commit Message

unbound: disable chroot to restore outbound forwarding capability